Water-Use Efficiency under Nutrient Addition Experiment along an Elevational Gradient

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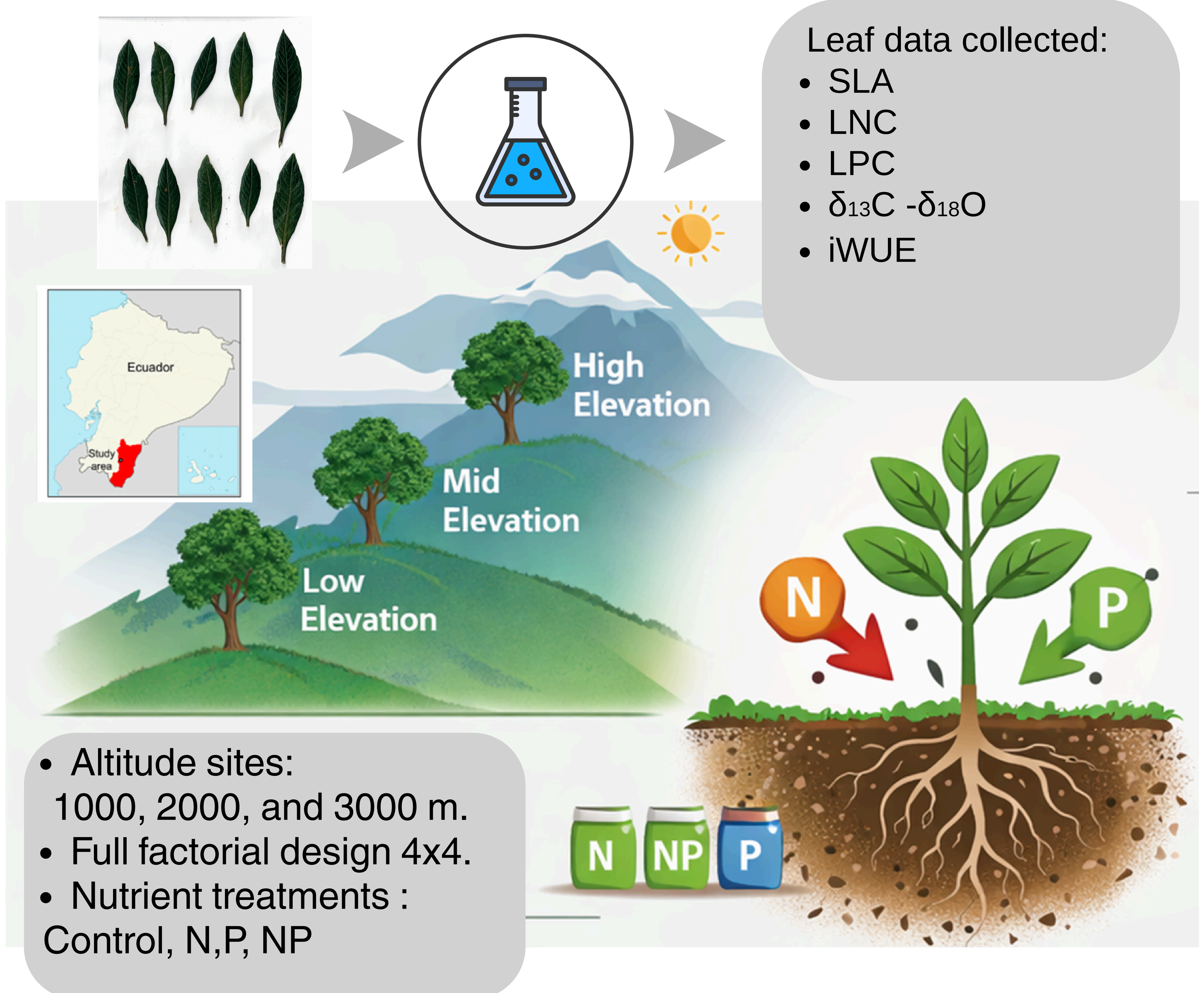
Background

- **Intrinsic water-use efficiency (iWUE)** is a key physiological parameter that reflects how plants balance carbon assimilation with stomatal regulation of water loss.
- **NUMEX** is a long-term nutrient manipulation experiment established in 2009 in a tropical montane forest in the South of Ecuador. It was designed to evaluate ecosystem responses to sustained, moderate additions of nitrogen (N) and phosphorus (P).
- There is **limited information** on the ecophysiological responses of tropical montane forests in the Andes in terms of iWUE.

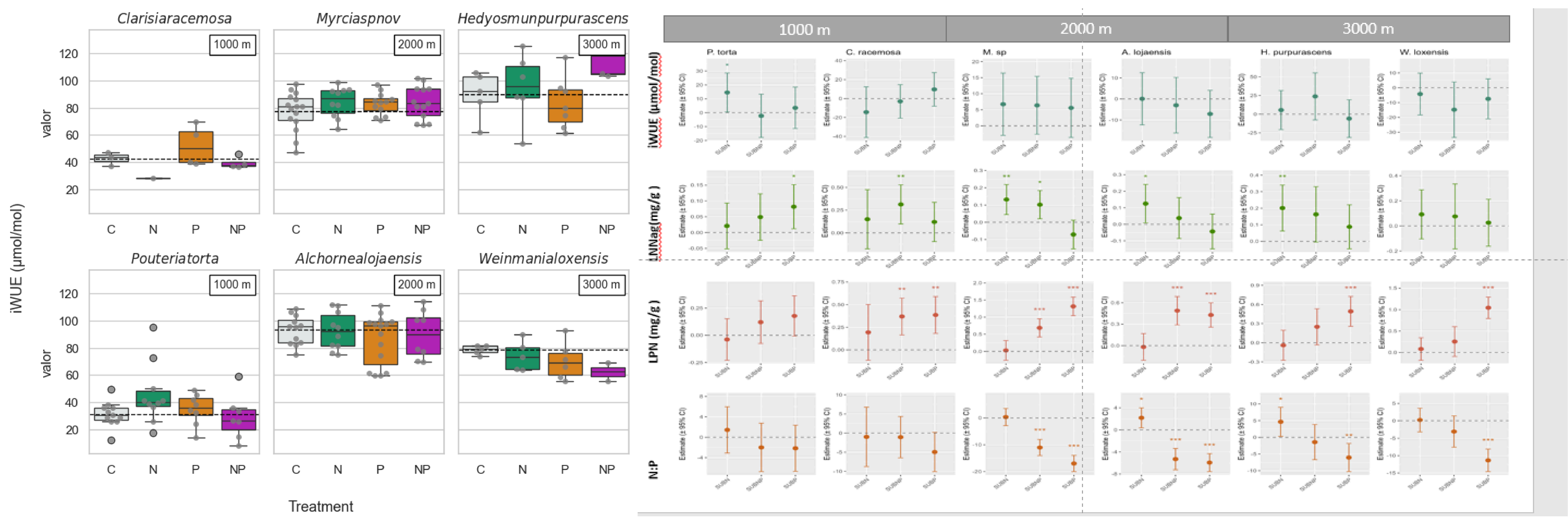
Objective

Evaluate the effect of moderate nutrient addition to l**eaf foliar nutrients and iWUE** after 15 years of fertilization across treatments in an elevational gradient.

Methodology



Results



Conclusions

- iWUE **no significant response** to those from the treatments compared to the ones of the control at all elevations.
- In general, **iWUE increased in montane species at higher elevations**. This pattern suggests adaptation to colder, drier conditions at the tops of the mountains.
- The results provide **new insights** into water-use strategies of **tropical montane tree species** and highlight the importance of **long-term nutrient manipulation experiments** for addressing knowledge gaps in tropical forest eco-physiology

Future work

- Exploration of more ecophysiological parameteres.

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